

# EXTERNAL\_INTELLIGENCE.md

## SynChain External Intelligence Layer

### Purpose

Traditional supply chain systems only use internal company data.

SynChain extends decision-making by incorporating external intelligence.

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### Objectives

Detect external events before they impact operations.

Examples:

- Chip shortages
  - Port closures
  - Transportation strikes
  - Commodity price increases
  - Regulatory changes
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### Intelligence Sources

#### Tavily Search

Purpose:

- Industry research
  - Supply chain news
  - Company-specific news
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#### GDELT

Purpose:

- Global event monitoring
- Economic events

- Geopolitical events
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## RSS Feeds

Examples:

- Logistics publications
  - Manufacturing publications
  - Technology industry sources
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## News Intelligence Agent

Responsibilities:

- Fetch articles
- Extract events
- Calculate impact scores

Output:

```
{ "event": "Semiconductor shortage", "impact_score": 92, "affected_industries": [ "Electronics",  
"Automotive" ] }
```

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## Market Intelligence Agent

Responsibilities:

- Detect trends
  - Analyze growth signals
  - Identify market shifts
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## Event Processing Pipeline

News Sources ↓ News Agent ↓ Event Extraction ↓ Impact Assessment ↓ Digital Twin Update ↓ Demand Agent ↓ Recommendation Engine

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## Intelligence Categories

- Manufacturing
  - Logistics
  - Transportation
  - Retail
  - Semiconductors
  - Commodities
  - Energy
  - Consumer Goods
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## Future Enhancements

- Social sentiment analysis
- Government policy tracking
- Commodity market integration
- Weather intelligence